S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: To apply Principal Component Analysis for dimensionality reduction and visualize the transformed feature space.

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Aim: To apply Principal Component Analysis for dimensionality reduction and visualize the transformed feature space.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To understand the concept of dimensionality reduction.
- To apply Principal Component Analysis (PCA) on a multivariate dataset.
- To reduce high-dimensional data into fewer principal components.
- To retain maximum variance while reducing the number of features.
- To visualize the transformed lower-dimensional feature space.

THEORY:

In many real-world applications, datasets contain a large number of features (high dimensional data). High-dimensional data can cause:

- Increased computational complexity
- Overfitting in machine learning models
- Difficulty in visualization
- Multicollinearity among features

To overcome these problems, dimensionality reduction techniques are used.

Principal Component Analysis (PCA) is one of the most widely used dimensionality reduction techniques.

Definition of PCA:

Principal Component Analysis (PCA) is an unsupervised statistical technique used to transform a large set of correlated variables into a smaller set of uncorrelated variables called principal components, while retaining most of the variance present in the dataset.

Objective of PCA:

The main objective of PCA is:

To maximize variance while minimizing dimensionality

PCA finds new axes (principal components) such that:

- First principal component (PC1) → captures maximum variance
- Second principal component (PC2) → captures maximum remaining variance
- And so on

Each new component is orthogonal (perpendicular) to the previous ones.

Mathematical Background:

Let the dataset be represented as:

X=[x1, x2, x3, ..., xn]

Where:

- n = number of observations
- p = number of features

Step 1: Data Standardization

Since PCA is variance-based, features must be standardized:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- μ = mean of feature
- σ = standard deviation

This ensures all features contribute equally.

Step 2: Covariance Matrix Calculation

The covariance matrix represents the relationship between features:

$$Cov(X) = \frac{1}{n-1}(X - \bar{X})^T(X - \bar{X})$$

If covariance is high → features are strongly related.

Step 3: Eigenvalues and Eigenvectors

From the covariance matrix, we solve:

$$|Cov - \lambda I| = 0$$

Where:

- λ = eigenvalues
- I = identity matrix

Eigenvectors represent directions of maximum variance. Eigenvalues represent the magnitude of variance along those directions.

Step 4: Explained Variance

Each principal component explains some portion of total variance:

$$\text{Explained Variance Ratio} = \frac{\lambda_i}{\sum \lambda}$$

Higher eigenvalue → more important component.

Step 5: Feature Transformation

Original data is projected onto new feature space:

$$Z = XW$$

Where:

- X = original standardized data
- W = matrix of selected eigenvectors
- Z = transformed lower-dimensional data

<u>Geometric Interpretation:</u>

PCA rotates the coordinate system such that:

- New axes align with maximum variance directions
- Data spread is maximized along new axes
- Dimensions with least variance are discarded

Thus, PCA finds the best-fit low-dimensional subspace.

Properties of Principal Components:

1. Principal components are orthogonal to each other
2. PC1 captures maximum variance
3. Total variance remains same after transformation
4. PCA is a linear transformation
5. Components are ranked in decreasing order of importance

Advantages of SVM

1. Reduces dimensionality effectively
2. Removes multicollinearity
3. Improves computational efficiency
4. Helps prevent overfitting
5. Useful for visualization (2D or 3D plots)
6. Retains most important information

Disadvantages of SVM:

1. Does not use class labels
2. Components are not easily interpretable
3. May discard useful low-variance features
4. Assumes linear relationships
5. Sensitive to scaling (must standardize data)

PROGRAM CODE:

OUTPUT (SCREENSHOT):

CONCLUSION:

DISCUSSION AND VIVA VOCE:

1. In a real-world machine learning pipeline, when would you apply PCA and when would you avoid it?
2. How do you decide the optimal number of principal components in an industry project?
3. Why is feature scaling mandatory before applying PCA in production systems?
4. In a healthcare diagnosis system, why might PCA reduce model interpretability?
5. How does PCA help in reducing overfitting in high-dimensional datasets?
6. Can PCA improve model training time in large-scale systems? Explain how.
7. Why is PCA not always suitable before tree-based models like Random Forest or XGBoost?
8. In image processing applications, how does PCA help in compression?
9. What are the limitations of PCA when dealing with non-linear data in industry?
10. How would you evaluate whether PCA actually improved your model performance in production?

REFERENCE

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